Forecasting California Electricity Demand Using Weather Features and Linear Regression

1. Project Overview and Problem Statement

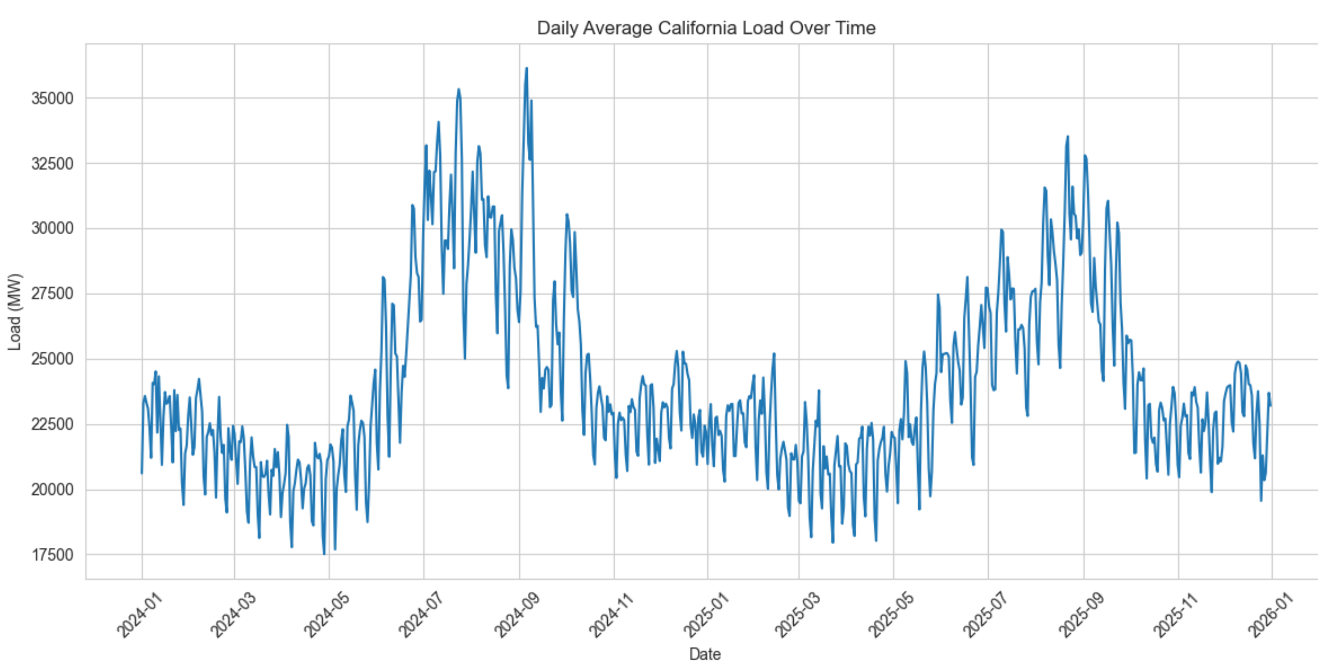
The primary objective of this project is to forecast short-term California electricity demand using daily weather and calendar-based features. Because electricity usage is strongly influenced by temperature, seasonality, and recent demand patterns, this project aims to build an interpretable model that captures those relationships while remaining simple enough to deploy in a lightweight forecasting app.

The end goal was not just to train a strong predictive model, but to build a full end-to-end workflow: collect and clean weather and demand data, engineer features, train and evaluate multiple regression models, and deploy the final model in a Streamlit app that generates a 4-day California demand forecast using live weather forecasts and recent CAISO (California Independent System Operator) load data.

2. Data Wrangling

In this project, I transformed raw weather and electricity demand data into a daily modeling dataset suitable for supervised regression.

- **Weather data source:** Hourly weather observations for 2024 and 2025 were collected from Meteostat for five major California cities: Los Angeles, San Francisco, San Diego, San Jose, and Fresno.
- **Weather feature selection:** From the full weather records, I kept a focused set of high-signal variables: temperature, relative humidity, precipitation, and wind speed.
- **Electricity demand data source:** CAISO demand data was collected through GridStatus. The raw CAISO demand data was first available at a higher frequency and then resampled to hourly averages before being aggregated to the daily level.
- **Daily aggregation:** Weather data was aggregated to daily means, maxima, minima, and sums where appropriate. Daily CAISO demand features were also constructed, including daily mean, max, min, and standard deviation.
- **Final merged dataset:** The final daily modeling table contains one row per day from January 1, 2024 through December 31, 2025, with weather summaries for all five cities and daily California load features. The target variable was defined as target_load_mw_mean, equal to the daily mean CAISO load.



3. Exploratory Data Analysis

Initial exploratory analysis showed that California electricity demand has a clear seasonal structure and a strong relationship with temperature.

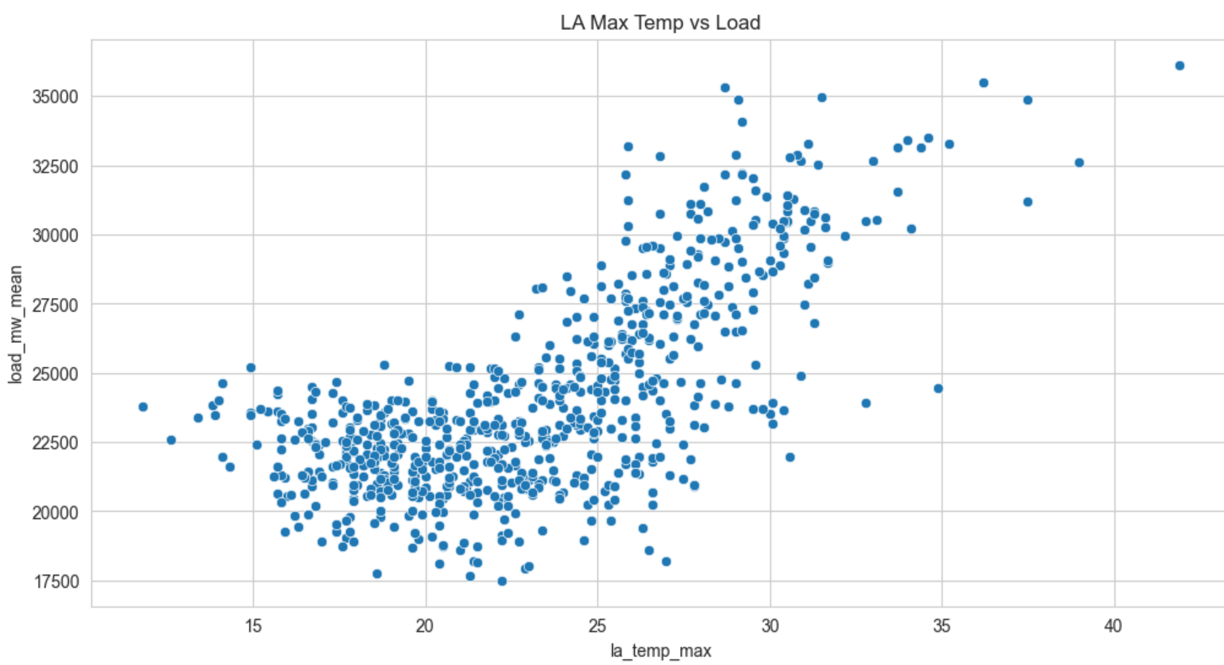
a) Demand Over Time

Daily California load shows broad seasonal variation across the two-year window. Demand tends to rise in warmer periods and fall during milder months, with visible medium-term fluctuations throughout the year. A 7-day rolling average makes the longer seasonal pattern easier to see and confirms that electricity demand is not random from day to day, but follows a combination of weather-driven and calendar-driven behavior.

b) Temperature and Demand

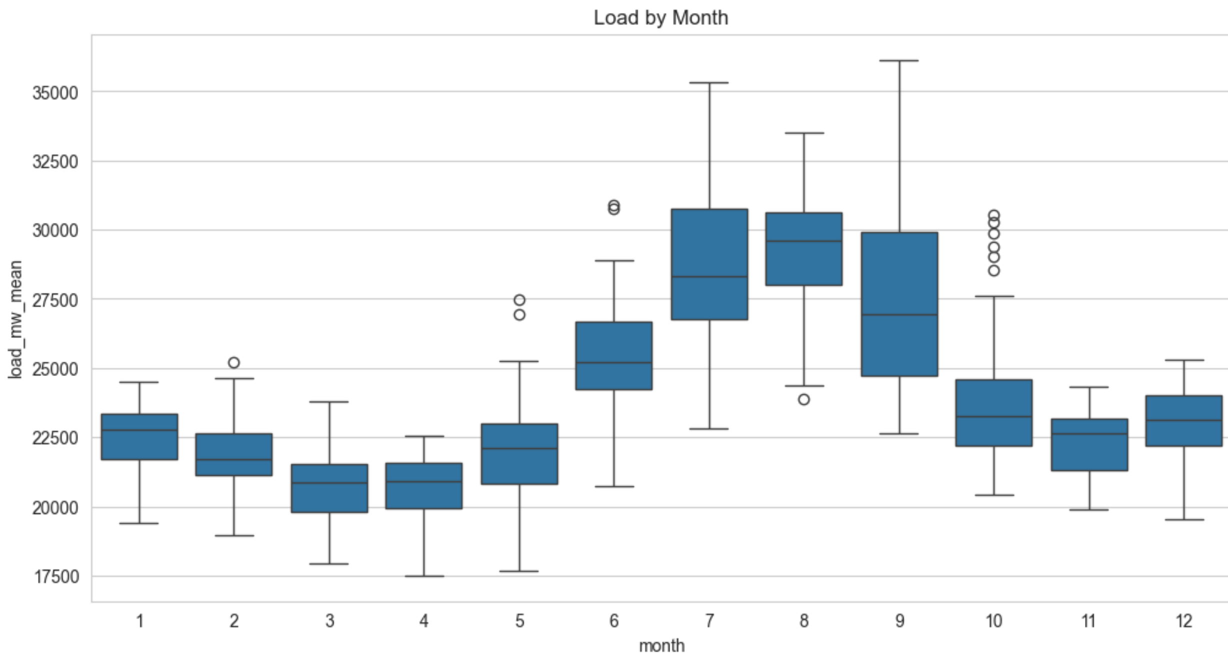
Temperature appears to be the most important weather driver. Scatterplots of load against maximum temperature, especially for Los Angeles and the aggregated cross-city temperature features, show that demand generally increases as temperatures rise. This is consistent with higher cooling demand during warmer periods.

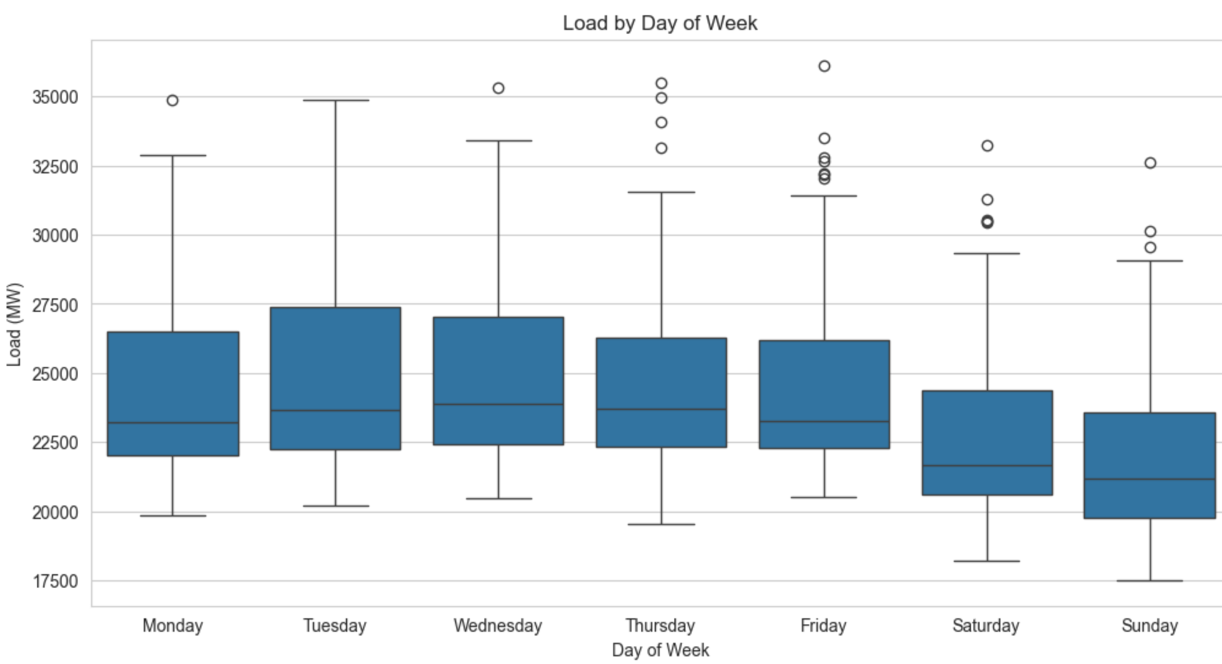
The relationship is not perfectly linear across the full range, which motivated the inclusion of engineered nonlinear temperature features such as cooling degree and heating degree variables during preprocessing. The EDA also suggested that aggregated multi-city weather features were more useful than relying on any single city alone.



c) Seasonality and Calendar Effects

Load also varies systematically by month and day of week. Monthly boxplots show strong seasonal differences, while weekday patterns suggest the model may benefit from explicit calendar features. These findings supported the inclusion of cyclical calendar encodings and weekend/season indicator variables during preprocessing.





4. Feature Engineering and Pre-Processing

To prepare the data for modeling, I engineered features designed to capture the major short-term drivers of electricity demand.

- **Lag feature:** I created lag1_load, defined as the previous day's target load. This allows the model to incorporate short-term persistence from one day to the next.
- **Aggregated weather features:** I averaged temperature, humidity, and wind features across the five cities and summed precipitation across cities to create statewide summary weather variables such as temp_mean_all, rhum_mean_all, prcp_sum_all, and wspd_mean_all. I also created temp_max_all, temp_min_all, and temp_range_all.
- **Nonlinear temperature features:** I engineered cooling_degree, heating_degree, squared versions of each, and binary indicators for hot and very hot days. These features help the model capture nonlinear weather-demand relationships.
- **Calendar features:** I added seasonal indicators (is_summer, is_winter, is_shoulder_season) and cyclical encodings for month, day of week, and day of year using sine and cosine transformations.
- **Interaction features:** To allow the model to represent more context-specific patterns, I added interactions such as hot-weekend, cold-weekend, summer-cooling, winter-heating, and humidity-heat terms.
- **Leakage prevention:** Raw same-day load summary variables were explicitly dropped before modeling to avoid leaking direct information about the target into the feature set.

The final modeling dataset was then split chronologically into train, validation, and test sets so that evaluation would reflect a real forecasting setting rather than a randomized split.

5. Modeling: Regression Forecasting

I evaluated four baseline regression models:

- Linear Regression
- Ridge Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Each model was wrapped in a Scikit-learn pipeline containing a preprocessing step and the estimator itself. Numeric features were standardized, and any categorical features would be one-hot encoded if present.

a) Model Selection and Evaluation

Models were compared on the validation set using MAE, RMSE, and R^2. Linear Regression achieved the best validation performance, with a validation MAE of about 565 MW, RMSE of about 732 MW, and R^2 of about 0.923. After retraining on the combined train and validation data, the final linear regression model achieved a test MAE of about 597 MW, test RMSE of about 765 MW, and test R^2 of about 0.887.

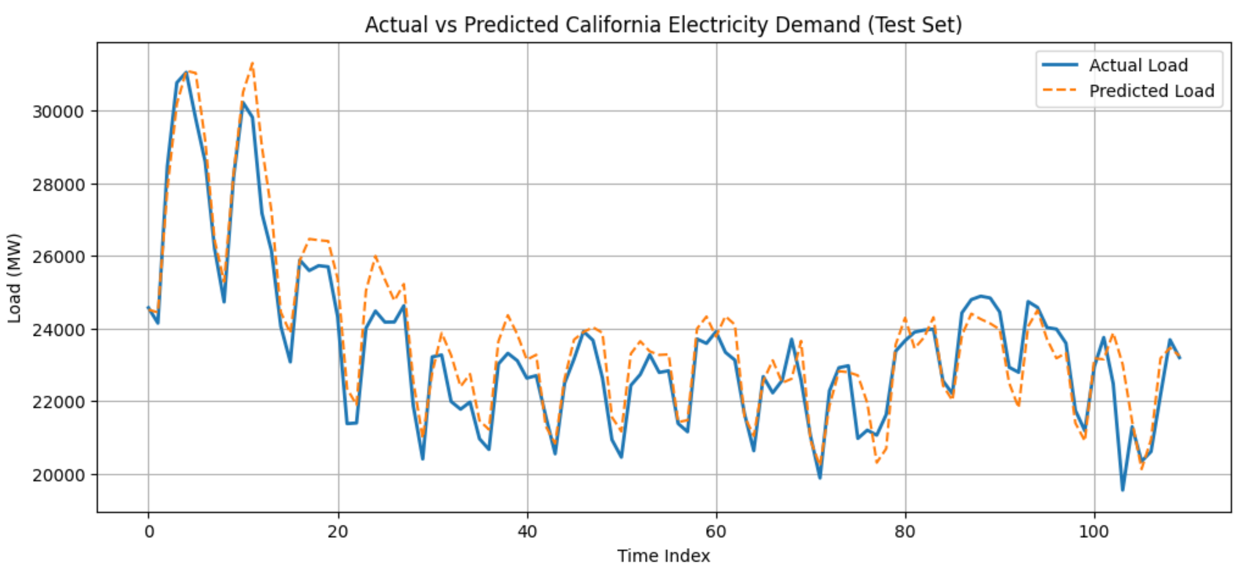
These results indicate that a relatively simple linear model was able to explain most of the variance in daily California electricity demand once the relevant weather, calendar, and lag features were engineered appropriately.

Decision: Linear Regression was selected as the final model because it performed best on validation RMSE and remained strong on the held-out test set, while also being the easiest model to interpret and deploy.

Model	Validation MAE	Validation RMSE	Validation R^2	Test MAE	Test RMSE	Test R^2
Linear Regression	564.93	731.58	0.9233	596.91	765.35	0.8874
Gradient Boosting	640.43	806.99	0.9067	—	—	—
Ridge	645.86	844.79	0.8977	—	—	—
Random Forest	778.70	941.07	0.8731	—	—	—

b) Prediction Quality

A comparison of actual versus predicted test values showed that the model generally tracked daily demand well, with errors usually small relative to the full range of load values. This supports the conclusion that temperature, seasonality, and recent load are enough to produce a strong short-term forecasting baseline.



6. Core Findings

The final model suggests several important takeaways about California electricity demand.

- **Temperature is the dominant driver.** Both EDA and model performance indicate that weather, especially temperature, explains a large share of daily demand variation.
- **Short-term persistence matters.** The lag1_load feature plays an important role, showing that yesterday's demand is highly informative for today's demand.
- **Simple models can work very well when feature engineering is strong.** Linear Regression outperformed the more flexible alternatives tested here, suggesting that the underlying relationships are structured enough to be captured without a highly complex model.

- **Aggregated statewide weather summaries are effective.** Combining weather across multiple major California cities gave the model a broader and more stable signal than relying on a single location.
- **Calendar context improves forecast quality.** Seasonal indicators, cyclical date encodings, and weekend/weather interactions all helped turn a basic regression problem into a stronger short-term forecasting system.

Overall, the project shows that California electricity demand is driven primarily by temperature, seasonal timing, and short-term demand persistence. With the right feature engineering, a simple linear model is able to capture these patterns effectively and generalize well to unseen data.

7. Streamlit Application and Future Work

The final model was deployed in a Streamlit app (predicting-ca-electricity-demand.streamlit.app) that forecasts California electricity demand for the next four days. The app pulls weather forecasts from OpenWeatherMap, retrieves recent CAISO load data through GridStatus, reconstructs the same engineered feature set used in training, and then generates recursive daily forecasts using the saved linear regression pipeline. The app also displays forecast cards, a line chart, and a summary table.

A key implementation lesson from deployment was the importance of training-inference consistency. During app development, forecast quality depended heavily on matching the same temperature units, lag logic, and engineered features used during training. Once those inputs were aligned, the app produced realistic daily demand forecasts in the correct scale.

Future work could improve the system in several ways. First, the model could be retrained on a longer historical period to improve robustness. Second, prediction intervals could be added so the app shows forecast uncertainty in addition to point forecasts. Finally, more advanced time-series or gradient-based models could be explored as follow-up baselines, although the current linear approach already performs strongly and offers the advantage of simplicity and interpretability.